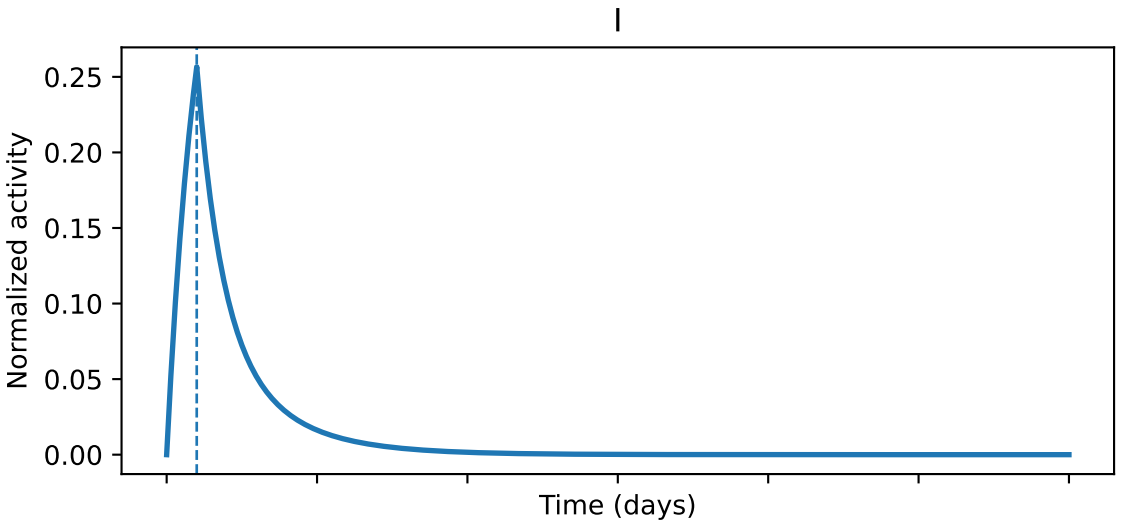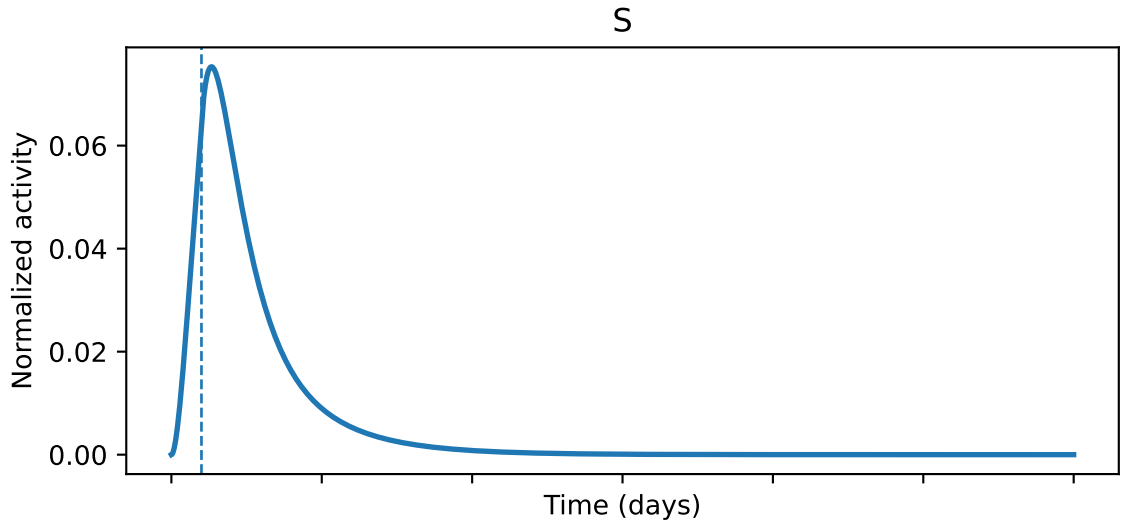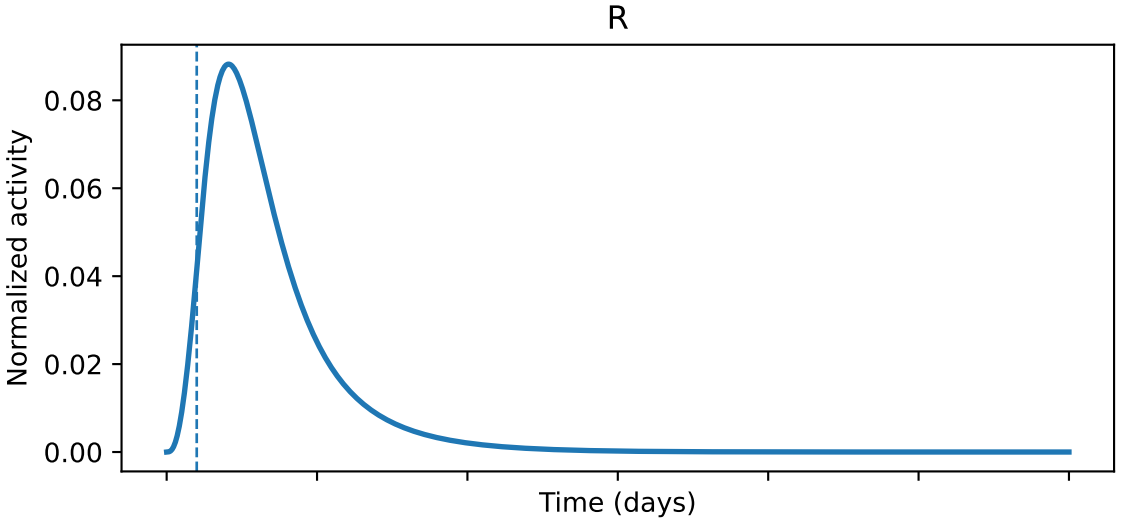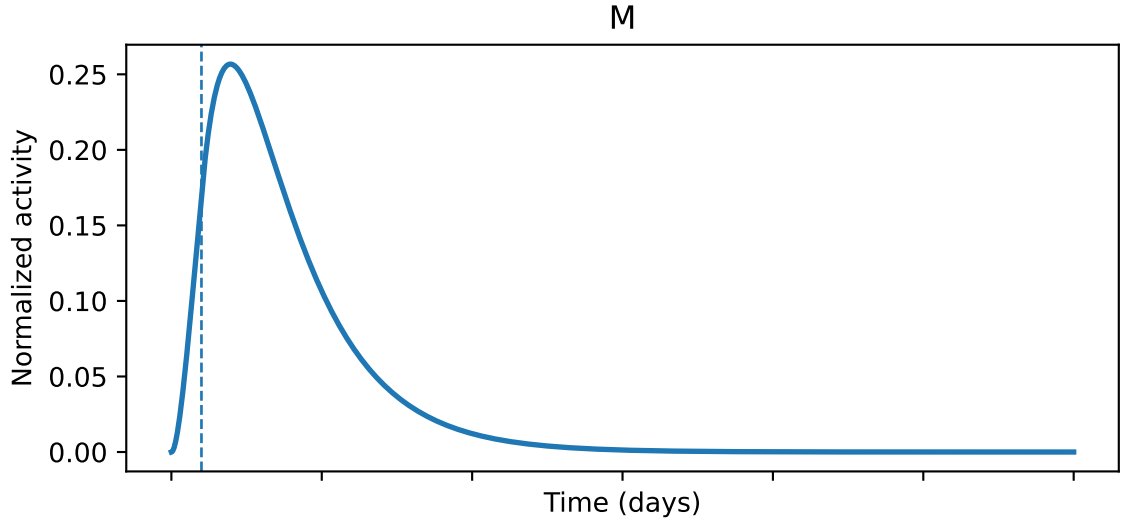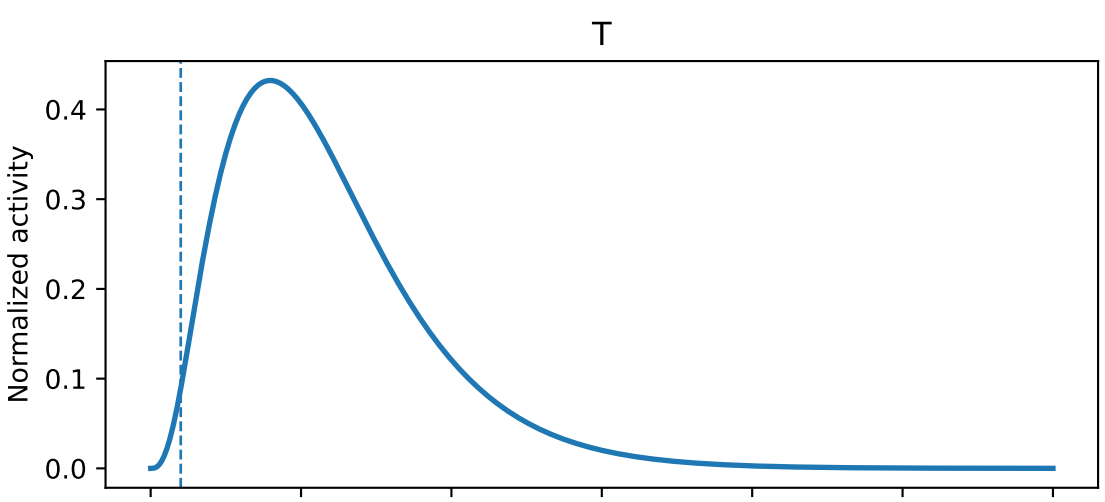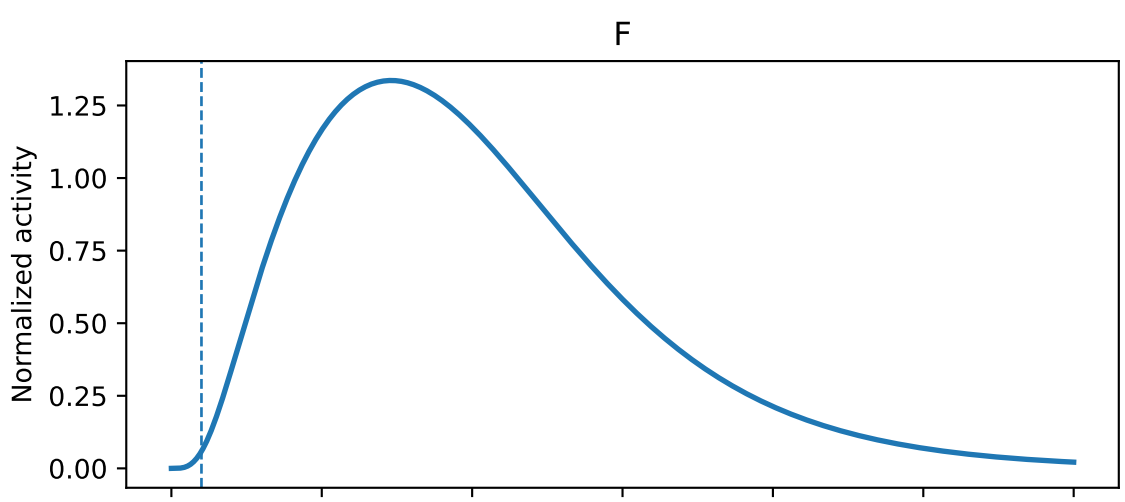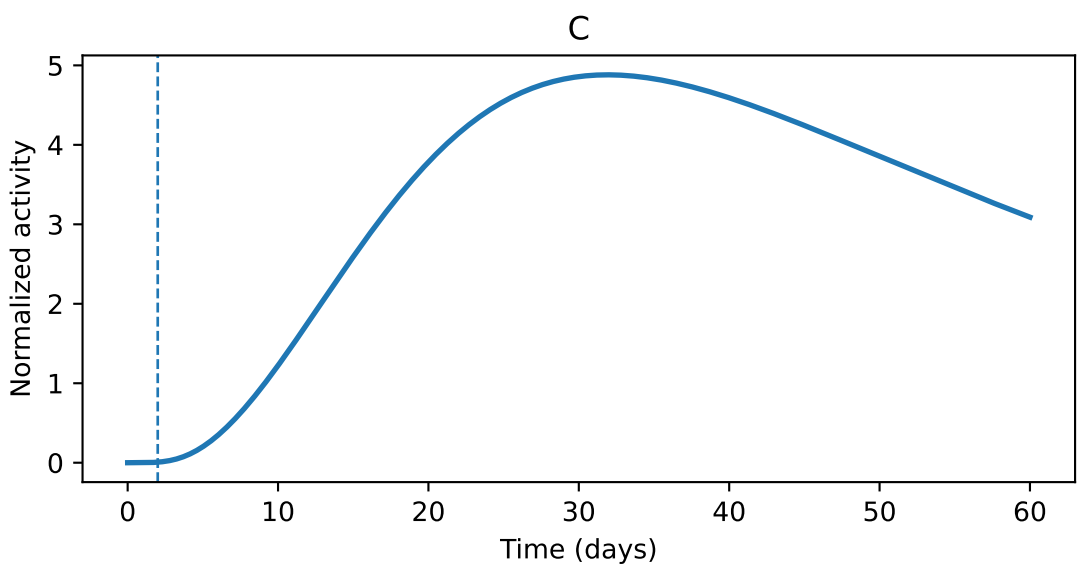Model A: Response to a Transient Inflammatory Input